

Latent Dimensions and Country Clusters in the Human Freedom Index

Digital Strategies for the Social Sciences · Linköping University · Patricia Givort Cruz Cabral

Introduction

The Human Freedom Index (HFI) is a dataset that measures human freedom across countries using indicators of personal, civil, and economic freedom. In the HFI framework, freedom is understood as the absence of coercive constrain, people should be able to make choices about their lives without arbitrary interference, as long as they respect the same rights of others (Vasquez and Porcnik, 2018).

The question of how personal and economic freedoms relate to each other has been debated in political theory for a long time. Classical liberal thinkers such as Friedman (1962) argued that economic freedom and civil liberties are fundamentally connected: in their view, a free-market system is a precondition for political freedom, and the two tend to reinforce each other. Sen (1999) challenged this idea, arguing that freedom must be understood as multidimensional, with social development, political rights, and civil liberties forming relatively independent dimensions that do not automatically follow from market liberalization. A more recent challenge to the classical liberal view comes from what Milanovic (2019) calls 'liberal meritocratic capitalism' versus 'political capitalism' illustrated by cases such as China or some Gulf states, where significant economic openness coexists with strict restrictions on civil liberties and political participation. These theoretical tensions suggest that the relationship between personal and economic freedom across countries is unlikely to be simple or linear, which is precisely why an exploratory analysis is useful here.

When I first saw the Human Freedom Index dataset, I found it interesting that freedom was not measured as a single concept, but divided into several personal, civil, and economic dimensions. This raised a question about how these dimensions relate to each other across countries. Do personal and economic freedoms tend to form one shared structure, where countries that score highly in one area also score highly in others? Or do they combine in different ways, creating distinct country profiles?

This question is especially relevant because a country may perform relatively well on some economic indicators while still having restrictions on personal or civil freedoms, or the opposite may occur. For this reason, the dataset is suitable for an exploratory analysis using PCA and clustering. PCA can show whether the selected freedom indicators are organized around common latent dimensions, while HCPC can identify whether countries with similar combinations of personal and economic freedom form recognizable geopolitical profiles.

The research question for this analysis is:

Do personal and economic freedoms form a single shared structure across countries, or do they combine into different geopolitical country profiles?

The aim of this exploratory question is to identify the main axes of variation across countries and to explore whether countries with similar freedom profiles form groups that reflect known geopolitical patterns. To answer

this question, Principal Component Analysis (PCA) is used to identify latent dimensions, followed by Hierarchical Clustering on Principal Components (HCPC) to group countries with similar freedom profiles.

Dataset and Variable Selection

The original HFI dataset contains observations from 2008 to 2016 and 123 variables. For this analysis, only the 2016 data is used. This choice makes the analysis cross-sectional: each row represents one country, which is the unit of analysis.

Eleven domain-level variables were selected: six personal freedom variables and five economic freedom variables. The personal freedom variables are: rule of law (pf_rol), security and safety (pf_ss), freedom of movement (pf_movement), religious freedom (pf_religion), freedom of expression (pf_expression), and identity and relationships (pf_identity). The economic freedom variables are: size of government (ef_government), legal system and property rights (ef_legal), sound money (ef_money), freedom to trade internationally (ef_trade), and regulation (ef_regulation).

Domain-level variables were chosen instead of the full set of detailed sub-indicators for two reasons. First, the detailed indicators contain more missing values, which would reduce the number of countries available for analysis. Second, domain-level variables are easier to interpret for a research question about broad freedom profiles. After removing countries with missing values, the final dataset contains 161 countries and 11 numerical variables.

Methods

PCA was applied because the selected variables are continuous numerical scores and the research question is about latent dimensions: underlying patterns that cannot be observed directly from individual variables. PCA reduces the 11 variables to a smaller number of components that capture the main axes of variation in the data.

All variables were standardized before the PCA (`scale.unit = TRUE` in `FactoMineR`). Standardization is necessary because PCA is sensitive to variable scale: without it, variables with larger numerical ranges would dominate the components regardless of their substantive importance. Standardizing ensures that all variables contribute on an equal footing.

After the PCA, Hierarchical Clustering on Principal Components (HCPC) was used to group countries according to their position in the PCA space. This combination is methodologically appropriate: PCA first summarizes the main structure of the data, and clustering then identifies groups of countries with similar profiles in that reduced space. The number of clusters was determined automatically by the HCPC algorithm based on the dendrogram structure (`nb.clust = -1`).

PCA Results

Variance Explained and Component Selection

The scree plot (Figure 1) shows the percentage of variance explained by each principal component. PC1 explains 49.2% of the total variance, PC2 explains 14.7%, and PC3 explains 10.7%. Together, the first three

components account for 74.6% of the total variation. Based on the elbow visible in the scree plot after the third component, and following the Kaiser criterion (eigenvalue > 1), three components were retained. The main interpretation focuses on PC1 and PC2, which together explain 63.9% of the variance and are sufficient to address the research question.

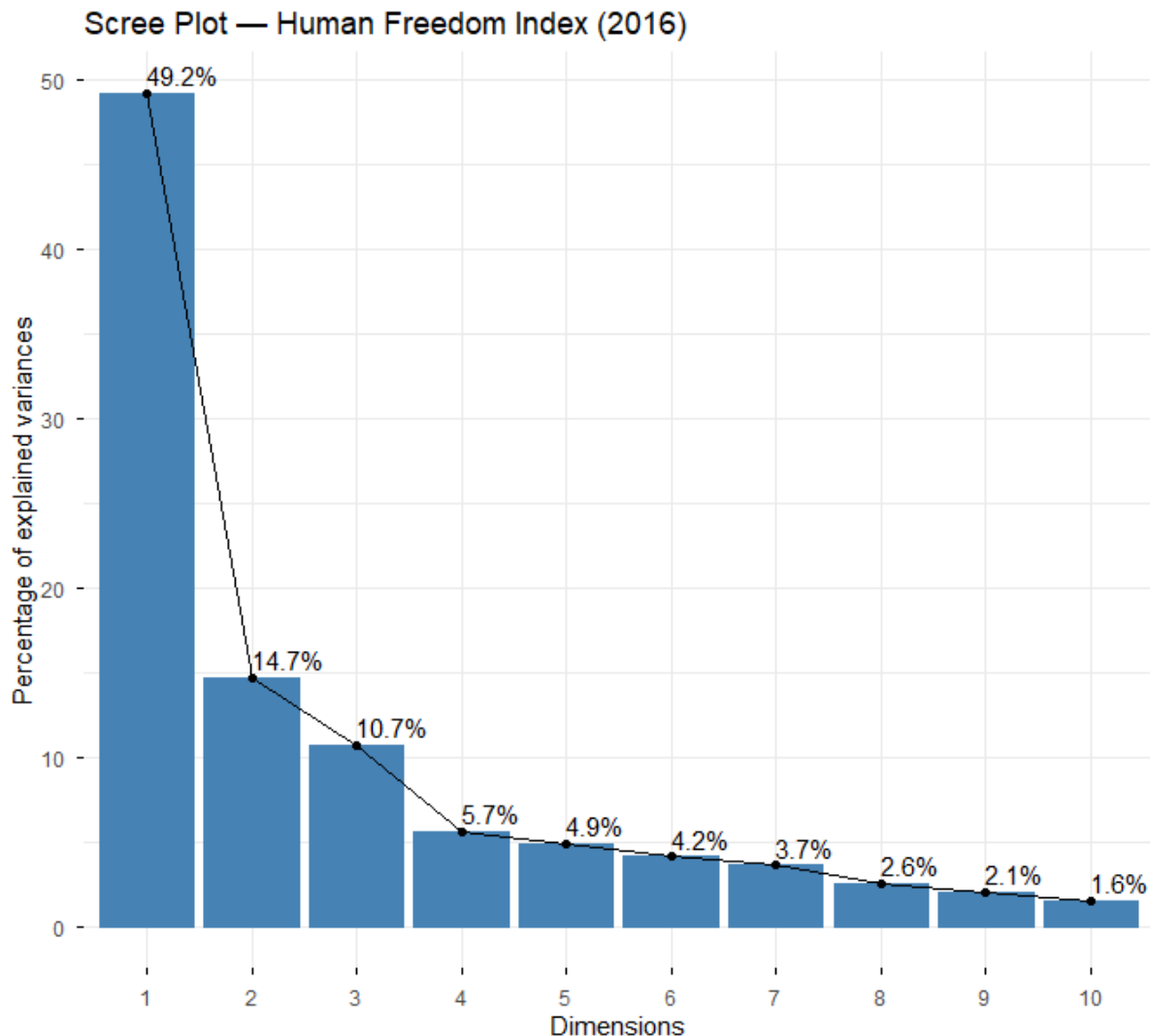


Figure 1. *Scree Plot — Human Freedom Index (2016). Percentage of variance explained by each principal component.*

Variable Factor Map and Interpretation of PC1

The variable factor map (Figure 2) shows how the 11 selected variables relate to the first two principal components. This is the most important plot for answering the research question, as it reveals which types of freedom move together across countries.

PC1 (49.2% of variance) is the dominant dimension. Variables loading strongly and positively on PC1 include rule of law (pf_rol = 0.88), legal system and property rights (ef_legal = 0.85), freedom to trade internationally (ef_trade = 0.85), security and safety (pf_ss = 0.77), freedom of expression (pf_expression = 0.74), regulation (ef_regulation = 0.67), and sound money (ef_money = 0.67). These variables appear on the lower right of the variable factor map and point in the same direction. This pattern indicates that countries with stronger rule of law also tend to have stronger legal systems, more trade freedom, greater security, and more stable economic

institutions. Critically, this dimension combines both personal freedom variables (pf_rol, pf_ss, pf_expression) and economic freedom variables (ef_legal, ef_trade, ef_money, ef_regulation). PC1 is therefore interpreted as a broad legal-institutional freedom dimension, reflecting a general level of freedom that is not limited to either personal or economic domains.

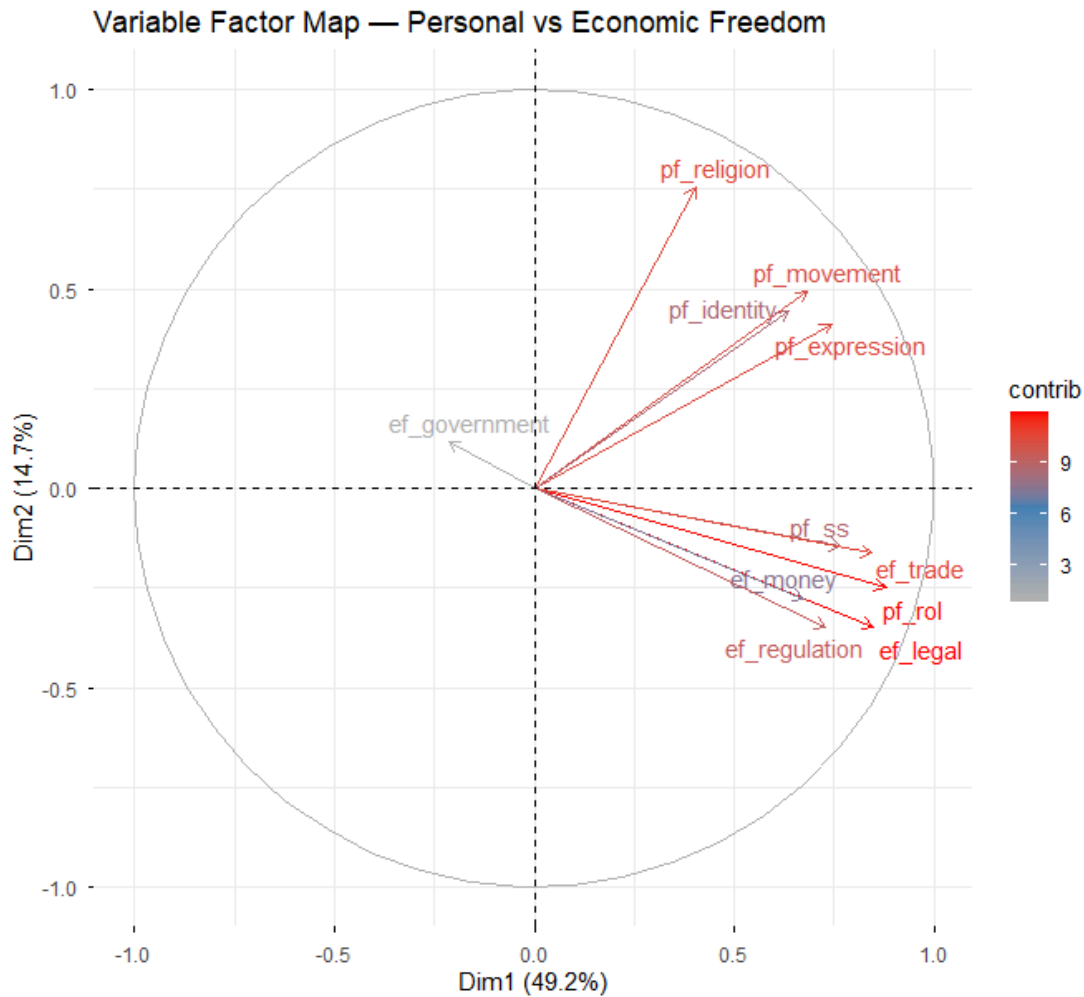


Figure 2. Variable Factor Map - Personal vs Economic Freedom. Variables are coloured by contribution to the PCA.

Interpretation of PC2

PC2 (14.7% of variance) captures a secondary distinction. Personal freedom variables related to religion (pf_religion = 0.76), movement (pf_movement = 0.49), identity (pf_identity = 0.44), and expression (pf_expression = 0.41) load positively in the upper part of the variable factor map, while economic-legal variables such as ef_legal load negatively. This suggests that PC2 reflects differences in the composition of freedom rather than the total level: some countries score higher on specific civil and personal liberties while having weaker economic-legal institutions, and vice versa. PC2 is interpreted as a dimension contrasting specific civil and personal liberties against legal-economic freedom.

The Role of Government Size

One notable finding is that ef_government (size of government) does not load strongly on either PC1 or PC2. In the variable factor map, it shows almost isolated in the left part of the plot with a short vector, indicating it is not well represented by the first two components. So government size behaves differently from the other economic

freedom variables, and does not consistently co-vary with either the total freedom profile or the personal-civil dimension. This shows that economic freedom is not internally uniform: larger government intervention is a partly independent dimension from the legal and trade aspects of economic freedom.

Clustering Results

After the PCA, HCPC identified four country clusters. The cluster structure is shown in Figure 3 and Figure 4, and the cluster profiles are summarized in Table 1.

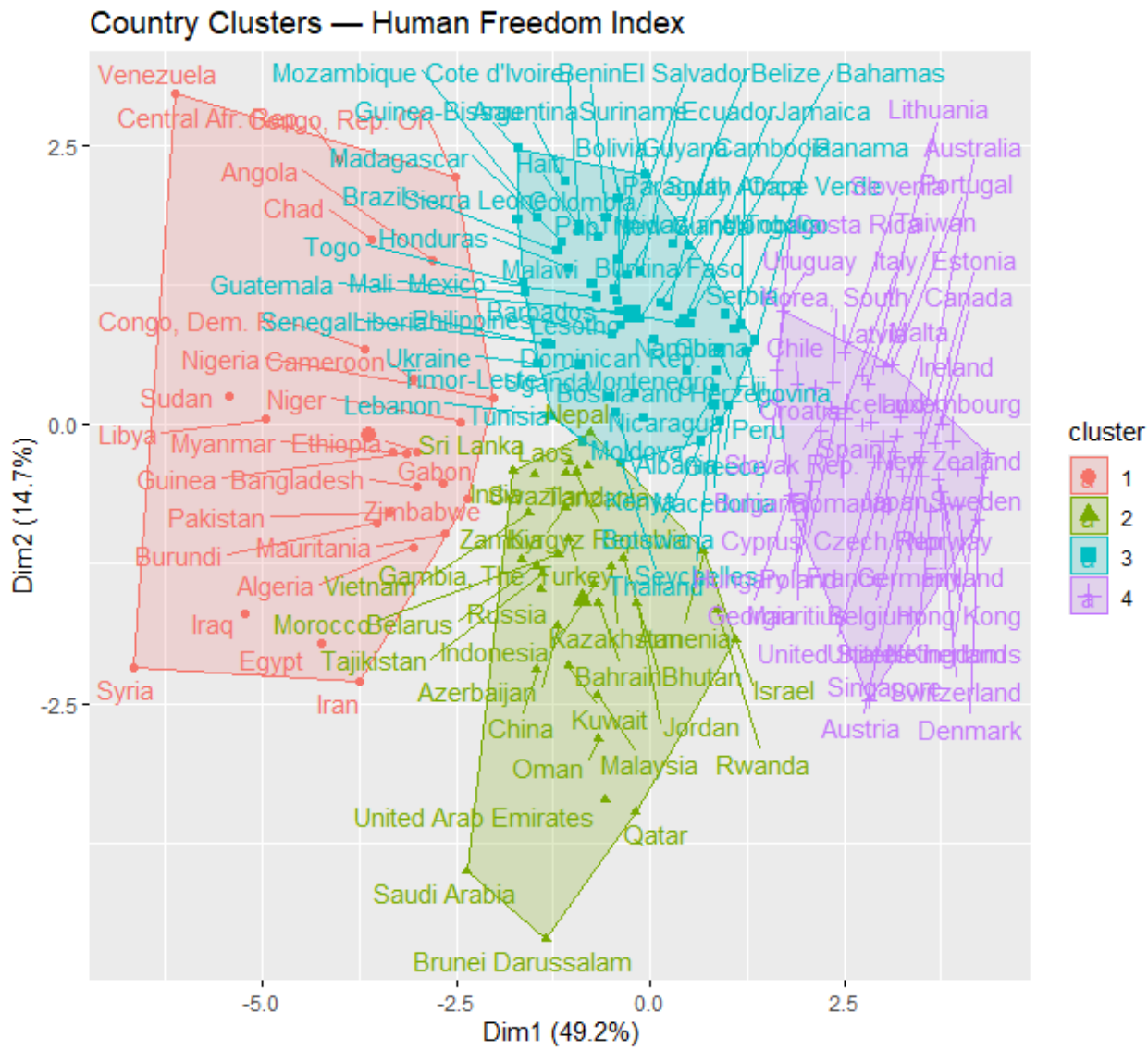


Figure 3. Country Clusters in PCA Space (2016). Countries are grouped by HCPC clustering. Colours indicate cluster membership; crosses mark cluster centroids.

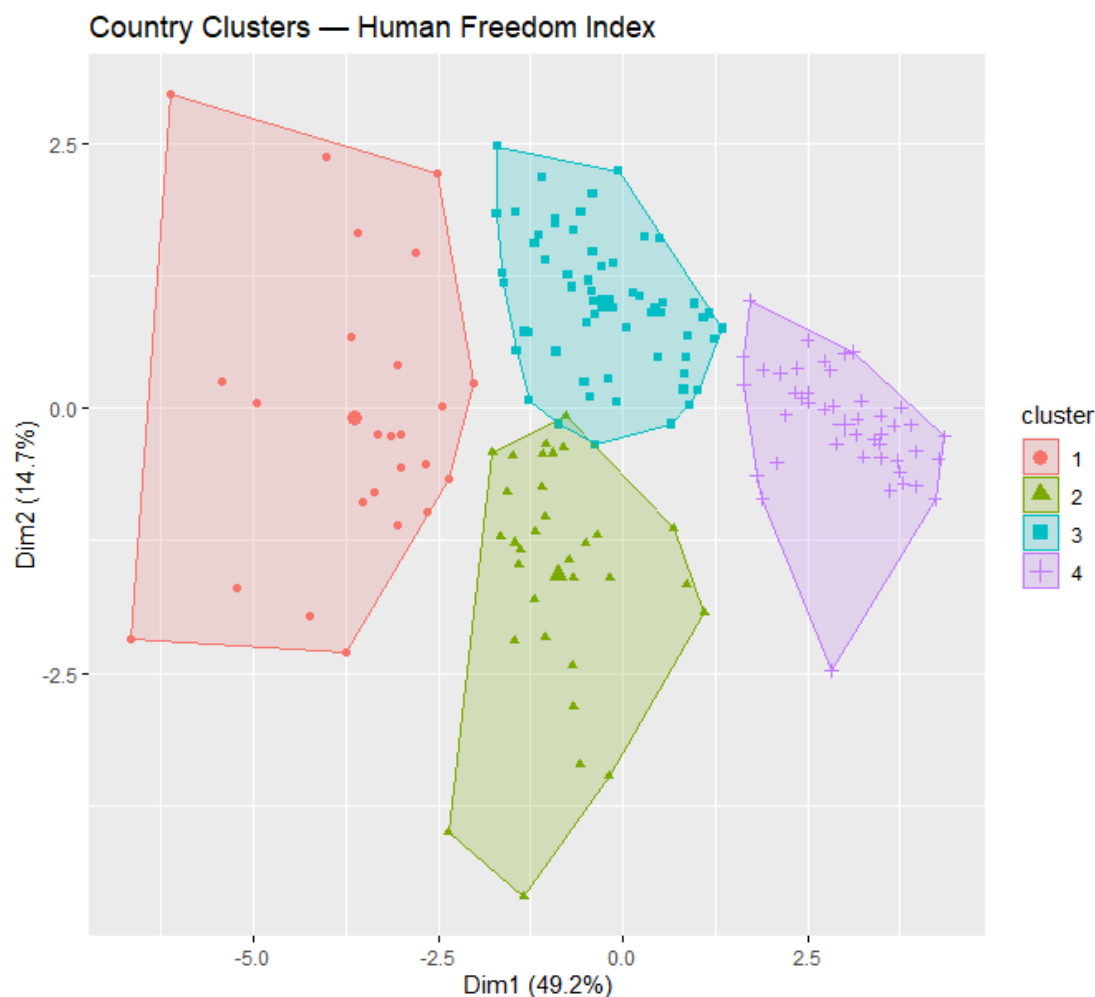


Figure 4. *Country Clusters in PCA Space (2016). Countries are grouped by HCPC clustering. Colours indicate cluster membership; crosses mark cluster centroids. Labels were removed for better visualization.*

Table 1. Summary of country clusters based on HCPC

Cluster	N	Freedom profile	Main regional pattern	Example countries
1	25	Lowest freedom profile: weak rule of law, movement, identity, legal system, and trade freedom.	Sub-Saharan Africa; Middle East & North Africa; South Asia.	Syria, Iran, Iraq, Venezuela, Zimbabwe.
2	32	Intermediate profile: stronger economic indicators, but weaker personal and civil freedoms.	Middle East & North Africa; South Asia; Caucasus & Central Asia.	China, Russia, Qatar, Saudi Arabia, UAE.
3	60	Mixed profile: stronger movement, religion, expression, and identity, but weaker legal institutions.	Latin America & Caribbean; Sub-Saharan Africa; Eastern Europe.	Brazil, Mexico, South Africa, Argentina, Ukraine.
4	44	Highest freedom profile: strong scores across most personal and economic freedom domains.	Western Europe; Eastern Europe; East Asia; North America; Oceania.	Sweden, Norway, Canada, Japan, New Zealand.

Table 1. *Summary of country clusters based on HCPC clustering of the Human Freedom Index (2016).*

Cluster 1 (N = 25) represents the lowest overall freedom profile in the analysis. Countries in this cluster, such as Syria, Iran, Iraq, Venezuela, Zimbabwe, Sudan, and Angola, show low scores in rule of law, movement, identity, legal system and property rights, and trade freedom. This cluster is predominantly composed of Sub-Saharan Africa (60%) and Middle East and North Africa (24%).

Cluster 2 (N = 32) shows an intermediate profile characterized by stronger economic indicators alongside more restricted personal and civil freedoms. Countries such as China, Russia, Qatar, Saudi Arabia, the United Arab Emirates, Malaysia, and Kazakhstan belong to this cluster. The cluster means show relatively high scores in sound money, regulation, trade, and security, but lower scores in religion, movement, identity, and rule of law. This pattern directly qualifies the PC1 finding: while personal and economic freedom tend to co-vary along the

main axis, Cluster 2 shows that some countries achieve reasonable economic freedom scores while maintaining significant restrictions on civil and personal liberties.

Cluster 3 (N = 60) is the largest cluster and displays a mixed freedom profile. Countries such as Brazil, Argentina, Mexico, Colombia, South Africa, Albania, Serbia, Ukraine, and the Philippines belong to this group. The cluster means show relatively high scores in movement, religion, expression, and identity, but weaker scores in rule of law and legal system and property rights. This suggests a profile in which personal freedoms are comparatively stronger, but institutional and legal conditions are weaker. The cluster includes countries from Latin America and the Caribbean, Sub-Saharan Africa, Eastern Europe, and South Asia.

Cluster 4 (N = 44) represents the highest overall freedom profile. Countries such as Norway, Sweden, Finland, Denmark, Switzerland, Canada, Australia, New Zealand, Japan, and Germany belong to this cluster. The cluster means show consistently high scores across most variables. This cluster is strongly associated with Western Europe (40.9%), but also includes countries from Eastern Europe, East Asia, North America, Oceania, and Latin America.

The regional distribution of the clusters shows recognizable geopolitical tendencies, but the clusters are not purely geographical. For example, Cluster 4 includes many Western European countries but also countries from other regions. Cluster 3 includes many Latin American and Sub-Saharan African countries alongside Eastern European nations. It confirms the PCA finding: geography matters, but does not fully determine freedom profiles.

A comprehensive examination of the Variable Factor Map (Figure 2) reveals that the strong vector convergence between indicators such as economic legal system (*ef_legal*) and the rule of law (*pf_rol*) along the positive axis of PC1 reflects a institutional reality: the legal security required for market operations (e.g., property rights protection, contract enforcement) is structurally bound to fundamental civil and criminal justice safeguards. So, PC1 consolidates into an axis of 'Structural Institutional Maturity.' Conversely, the isolated, nearly orthogonal trajectory of the size of government variable (*ef_government*) relative to personal freedom indicators demonstrates that the fiscal volume or tax burden of a state does not deterministically dictate the level of physical safety or civil rights enjoyed by its populace.

This structural independence is visibly reflected in the geopolitical composition of the clusters generated by the HCPC, which deconstruct the notion of a uniform, linear progression toward freedom. Cluster 1 represents a total systemic collapse across both dimensions, characteristic of failed states or severe authoritarian regimes. Conversely Clusters 2 and 3 reveal asymmetrical developmental trajectories. Cluster 2 embodies the phenomenon of 'authoritarian mismatch': these nations have successfully implemented market-oriented reforms integrated into global financial flows while maintaining heavily centralized state apparatuses that enforce restrictive controls over speech, association, and identity. Cluster 3 displays the inverse asymmetry: these societies function as open democracies with high levels of personal liberty, yet they face severe structural bottlenecks along PC1, manifested as inefficient judicial systems, chronic urban violence, and regulatory instability. Finally, Cluster 4 illustrates a high-synergy profile, demonstrating that robust, transparent legal institutions serve as a dual foundation for both highly dynamic market economies and comprehensive civil protections.

Limitations

This analysis has several limitations. It uses only 2016 data, so it does not capture how freedom profiles have changed over time. Repeating the PCA for earlier years, such as 2008, would show whether the component structure is stable across the period covered by the dataset.

Second, the results depend on the choice of variables. Using all detailed sub-indicators instead of domain-level summaries could produce a different PCA structure. Although, this would also increase missing data substantially and make interpretation more difficult the selected domain-level variables represent a deliberate trade-off between coverage and interpretability.

There are also technical limitations that come with the methods themselves meaning that PCA assumes that relationships between variables are linear. If countries' freedom profiles involve non-linear patterns or threshold effects, for example, where small changes in rule of law only matter above a certain level, PCA may not capture those dynamics accurately. The method is also sensitive to outliers, which can pull the principal axes in ways that affect the interpretation of all other countries. By retaining only PC1 and PC2 for the main analysis, approximately 36% of the total variance is set aside, which means some secondary patterns are not discussed.

Also, the Human Freedom Index reflects the methodological and ideological perspective of its publishers, the Cato Institute and the Fraser Institute, which emphasize a specific conception of freedom rooted in classical liberal principles. This means the dataset may not capture all dimensions of freedom equally, and the results should be read with this context in mind.

Conclusion

This analysis shows that personal and economic freedoms do not appear as completely separate dimensions in the Human Freedom Index. The PCA results indicate that the strongest structure in the data is a broad legal-institutional freedom dimension (PC1, 49.2% of variance), which combines variables from both personal and economic freedom including rule of law, legal system and property rights, trade freedom, security and safety, expression, regulation, and sound money. The co-movement of these variables suggests that, across countries, institutional conditions tend to support both personal and economic freedoms at the same time, rather than separating them.

Although, the results also show that this shared structure does not explain everything. PC2 captures a secondary distinction related to specific civil and personal liberties especially religion, movement, identity, and expression, which can vary independently from the legal-economic profile. In addition, government size behaves as a largely independent dimension from the other economic freedom variables. Freedom is therefore multidimensional: its primary empirical pattern is shared across personal and economic domains, but important differences in composition remain.

The clustering results make these differences more visible. HCPC identified four country profiles. Cluster 4 illustrates the high-synergy case where strong legal institutions support both market economies and civil protections. Cluster 1 shows the opposite: a general collapse of institutional conditions across both domains. Clusters 2 and 3 reveal two different types of asymmetry. Cluster 2 which includes China, Russia, and several Gulf states: shows that economic openness and market integration can coexist with systematic restrictions on

civil and political life, consistent with Milanovic's (2019) description of political capitalism. Cluster 3 which includes many Latin American and Eastern European countries shows the inverse: open democratic societies with meaningful civil freedoms that still face significant institutional and legal weaknesses.

The answer to the research question is that personal and economic freedoms form a strong shared structure across countries, but they also combine into different country profiles. These profiles show recognizable geopolitical patterns without being reducible to geography alone. The analysis supports the view that human freedom is genuinely multidimensional (Sen, 1999), while also confirming that a broad legal-institutional dimension combining civil and economic indicators is the dominant axis along which countries vary.

References

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<https://www.cato.org/human-freedom-index/2016>

Vásquez, I., & Porčnik, T. (2018). *The human freedom index 2018: A global measurement of personal, civil, and economic freedom*. Cato Institute, Fraser Institute, and Friedrich Naumann Foundation for Freedom.
<https://www.cato.org/human-freedom-index/2018>

Appendix: R Code

The following R code was used for all analyses in this paper.

```
# =====  
# Assignment 2 – Human Freedom Index  
#  
# EDA methods: PCA + Clustering (HCPC)  
# =====  
  
# — Packages —————  
install.packages(c("FactoMineR", "factoextra", "tidyverse", "tibble", "gt", "webshot2"))  
library(FactoMineR) # PCA and HCPC  
library(factoextra) # PCA and cluster visualisations  
library(tidyverse) # data manipulation  
library(tibble)  
library(gt) # cluster summary table  
library(webshot2) # save gt table as PNG  
  
# — Load data —————  
hfi <- read.csv("hfi2008_2016.csv", stringsAsFactors = FALSE)  
dim(hfi) # 1458 rows, 123 columns
```

```

# — Filter to 2016 (cross-sectional: one row per country) ———
hfi_2016 <- hfi %>% filter(year == 2016)
nrow(hfi_2016) # 162 countries

# — Select 11 domain-level variables —————
# pf_ = personal freedom | ef_ = economic freedom
hfi_pca_data <- hfi_2016 %>%
  select(countries, region,
    pf_rol, # rule of law
    pf_ss, # security and safety
    pf_movement, # freedom of movement
    pf_religion, # freedom of religion
    pf_expression, # freedom of expression
    pf_identity, # identity and relationships
    ef_government, # size of government
    ef_legal, # legal system and property rights
    ef_money, # sound money
    ef_trade, # freedom to trade internationally
    ef_regulation # regulation
  )

# — Remove NAs, set country names as rownames —————
hfi_clean <- hfi_pca_data %>% filter(complete.cases(.))
nrow(hfi_clean) # 161 countries after removing missing values
country_names <- hfi_clean$countries
country_regions <- hfi_clean$region
hfi_num <- hfi_clean %>% select(-countries, -region)
rownames(hfi_num) <- country_names

# — PCA —————
# scale.unit=TRUE: standardizes variables – obligatory when scales differ
# graph=FALSE: suppress automatic plots; use fviz_* for control
pca_res <- PCA(hfi_num, scale.unit = TRUE, graph = FALSE)
pca_res$eig # eigenvalues and % variance explained
summary(pca_res) # full PCA summary

# — Figure 1: Scree plot —————
fviz_eig(pca_res, addlabels=TRUE, ncp=10) +
  ggtitle("Scree Plot – Human Freedom Index (2016)")

# — Figure 2: Variable factor map (correlation circle) —————
fviz_pca_var(pca_res, col.var="contrib",
  gradient.cols=c("grey70","steelblue","red"), repel=TRUE) +
  ggtitle("Variable Factor Map – Personal vs Economic Freedom")

# — Contribution plots (Appendix) —————
fviz_contrib(pca_res, choice="var", axes=1, top=11) + ggtitle("Contributions to PC1")
fviz_contrib(pca_res, choice="var", axes=2, top=11) + ggtitle("Contributions to PC2")

# — Figure 3: Countries by region —————
region_factor <- as.factor(country_regions)
fviz_pca_ind(pca_res, label="none", habillage=region_factor,
  addEllipses=TRUE, ellipse.level=0.68) +
  ggtitle("Countries by Region in PCA Space")

```

```

# — Loadings: which variables define each component —————
loadings <- as.data.frame(pca_res$var$coord)
loadings$freedom_type <- ifelse(grepl('^pf_', rownames(loadings)),
  "Personal Freedom", "Economic Freedom")
print(round(loadings[, c('Dim.1', 'Dim.2', 'freedom_type')], 3))

# — Clustering (HCPC) —————
# HCPC clusters countries in PCA space
# nb.clust=-1: automatic cluster number from dendrogram
set.seed(123)
hcpc_res <- HCPC(pca_res, nb.clust=-1, graph=FALSE)

# — Figure 4: Cluster plot —————
fviz_cluster(hcpc_res, repel=TRUE, show.clust.cent=TRUE) +
  ggtitle("Country Clusters – Human Freedom Index")

# — Cluster membership —————
cluster_data <- hcpc_res$data.clust %>%
  rownames_to_column('country') %>%
  left_join(hfi_clean %>% select(countries, region),
    by=c('country'='countries')) %>%
  select(country, region, clust) %>% arrange(clust)
print(cluster_data)

# — Cluster means —————
hfi_clean %>% mutate(clust=hcpc_res$data.clust$clust) %>%
  group_by(clust) %>%
  summarise(across(where(is.numeric), ~round(mean(.x, na.rm=TRUE), 2)))

# — Table 1: cluster summary (saved as PNG via gt) —————
cluster_summary_table <- tibble(
  Cluster = c(1,2,3,4), N = c(25,32,60,44),
  `Freedom profile` = c(
    "Lowest freedom profile: weak rule of law, movement, legal system, trade.",
    "Intermediate: stronger economic indicators, weaker personal freedoms.",
    "Mixed: stronger personal freedoms, weaker legal institutions.",
    "Highest freedom profile: strong scores across most domains."
  ),
  `Main regions` = c(
    "Sub-Saharan Africa; MENA; South Asia",
    "MENA; South Asia; Caucasus/Central Asia",
    "Latin America; Sub-Saharan Africa; Eastern Europe",
    "Western Europe; Eastern Europe; East Asia; N. America; Oceania"
  ),
  `Example countries` = c(
    "Syria, Iran, Iraq, Venezuela, Zimbabwe",
    "China, Russia, Qatar, Saudi Arabia, UAE",
    "Brazil, Mexico, South Africa, Argentina, Ukraine",
    "Sweden, Norway, Canada, Japan, New Zealand"
  )
)

cluster_table_gt <- cluster_summary_table %>% gt::gt() %>%
  gt::tab_header(title=gt::md("***Table 1. Summary of country clusters based on HCPC***"))
gt::gtsave(cluster_table_gt, 'cluster_summary_table_HCPC.png',
  expand=10, vwidth=1300, vheight=750)

```

```
# — Save figures —————  
# ggsave("scree_plot.png", width=8, height=5, dpi=300)  
ggsave("variable_factor_map.png", width=8, height=6, dpi=300)  
ggsave("regional_map.png", width=9, height=6, dpi=300)  
ggsave("country_clusters.png", width=8, height=6, dpi=300)
```